# Turning the Ratchet: Dynamic Screening with Multiple Agents

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# The Ratchet Effect

The ratchet effect impairs information revelation in dynamic relationships.

- An agent with a high performance today will tomorrow face a demanding incentive scheme.
- He should thus be reluctant to convey favorable information early in the relationship (Laffont and Tirole 1988).

# Main Results

This paper studies a dynamic contracting problem with a continuum of agents and limited commitment.

- A principal screens efficient agents using one-period contracts, but may revise contracts upon knowing an agent's type.
- Alterations of contracts are observable.

Specifically, this paper

1. Characterizes optimal contracts within a natural class;
2. Provides a necessary and sufficient condition under which information revelation can be fostered.

# Players and Types

A principal (she) interacts with a continuum of agents (he) of measure 1.

- Time discrete and infinite
- Shared discount factor $\delta \in (0, 1)$

Each agent has a type $\theta \in \{L, H\}$, affecting production cost.

- Private information
- The measure of low-cost type $\alpha_0 \in (0, 1)$

# Agent Payoff and Technology

In period $t$, given work quantity $q_t$ and payment $x_t$, a type-$\theta$ agent's payoff is

$$x_t - C_\theta(q_t),$$

where $C_\theta(\cdot)$ is the production cost function.

- Quantity $q_t \in [0, \overline{q}]$
- $C_\theta(\cdot)$ smooth, strictly increasing, and convex
- $C_\theta(0) = 0$, and $C'_H(q) > C'_L(q), \forall q \in [0, \overline{q}]$
- $\Delta C(\cdot) = C_H(\cdot) - C_L(\cdot)$ nonnegative and strictly increasing
- $\Delta C$ concave[1]

[1] Results still hold qualitatively without this assumption.

# Principal's Payoff and Social Surplus

The principal's payoff from an agent is

$$v(q_t) - x_t,$$

where $v(\cdot)$ is strictly increasing, strictly concave, and differentiable; $v(0) = 0$.

The social surplus from from type $\theta$ producing $q$ is

$$\pi_\theta(q) := v(q) - C_\theta(q).$$

For each $\theta$, define surplus-maximizing quantity $q^*(\theta) := \arg\max_q \pi_\theta(q)$.

- Interior solutions assumed: $0 < \pi_H(q_H^*) < \pi_L(q_L^*)$ and $0 < q_H^* < q_L^* < \bar{q}$.
- Agent's outside option (irreversible exit) is 0.

# Assumption on Type Distribution

**Assumption 1**

$$\alpha_0 < \frac{\pi_H(q_H^*)}{\pi_L(q_L^*)}$$

This ensures that the principal should not ignore the high types and extract full surplus only from the low types.

# The Contracting Environment

Three key features:

1. The principal can only commit to one-period contracts and may renege on her past promises.
   - However, any reneging is observable to all agents and could be punished.
2. Each agent has private information about his type, and any information revelation is voluntary.
3. Each agent can irreversibly leave the relationship in any period by refusing the current contract.

# Limited Commitment and Mechanisms

**Limited commitment**

The principal can

- Specify a contract for the current period (binding and enforceable once accepted);
- Make promises about what contracts to offer in future periods (able to renege on).

**Sequential revelation mechanisms**

Focus on this natural class of mechanism.[2]

[2] Whether or not the restriction to this class of mechanisms is without loss remains an open question. The revelation principle in Doval and Skreta (2022) does not extend directly to this setting with a continuum of agents.

# Contract Proposals in Period $t$

In period $t$, the principal proposes:

- A pooling contract $(q_t^P, x_t^P)$ for agents not having revealed their type.
- A sequence of separating contracts $\{(q_\tau^S(t), x_\tau^S(t))\}_{\tau \geq t}$ for agents revealing their type at $t$.
  - The revealing agent will be given the contract for period $t$ and promised the contracts for all $\tau > t$.
  - The sequence may depend on the revelation time $t$.

The principal can also freely propose a contract $(\widetilde{q}_t^S(T), \widetilde{x}_t^S(T))$ to agents who revealed at each $T < t$.

- This contract may differ from $(q_t^S(T), x_t^S(T))$, the one she promised.

# Agent's Choice Set

An agent's choice set in each period depends on whether he has revealed.

- For one who has not revealed his type:
  1. Continuing to take the pooling contract
  2. Starting to take the separating contract
  3. Quitting
- For one who announced his low type at some $T < t$:
  1. Taking the separating contract $(q_t^S(T), x_t^S(T))$
  2. Quitting

# Timeline

Timeline:

1. The principal proposes
   - A pooling contract $(q_t^P, x_t^P)$;
   - A sequence of separating contracts $\left\{(q_\tau^S(t), x_\tau^S(t))\right\}_{\tau \geq t}$ for those who wish to reveal at $t$;
   - $(\widetilde{q}_t^S(T), \widetilde{x}_t^S(T))$ for each $T < t$.
2. Each agent acts depending on whether he has revealed his low type.
3. The chosen contracts are enforced.

# Allocations

An allocation $\{(q_t^i, x_t^i)\}_{t\geq 0, i\in[0,1]}$ describes the work and compensation of each agent in every period.

- In a sequential revelation mechanism, an allocation can be rewritten as:

$$\{(q_t^P, x_t^P), (q_\tau^S(T), x_\tau^S(T)), r_t\}_{t\geq 0, \tau\geq t}$$

  where $r_t$ is the measure of low-cost agents who reveal at $t$.

- $R_t := \sum_{\tau<t} r_\tau$

# Continuation Payoffs and Profits

Continuation Payoffs

- Type-$\theta$ agent at $t$ (not yet revealed):

$$U^P_{\theta,t} := (1-\delta)\sum_{\tau=t}^{\infty}\delta^{\tau-t}[x^P_\tau - C_\theta(q^P_\tau)]$$

- Low-type agent having revealed at $T \leq t$:

$$U^S_{L,t} := (1-\delta)\sum_{\tau=t}^{\infty}\delta^{\tau-t}[x^S_\tau(T) - C_L(q^S_\tau(T))]$$

The continuation profit of the principal is

$$\begin{aligned}\Pi(t) :=&(1-\delta)\sum_{\tau=t}^{\infty}\delta^{\tau-t}[(1-\alpha_0)+(\alpha_0 - R_\tau - r_\tau)][v(q^P_\tau) - x^P_\tau] \\ &+(1-\delta)\sum_{\tau=t}^{\infty}\delta^{\tau-t}\left[\sum_{T=0}^{\tau} r_T[v(q^S_\tau(T)) - x^S_\tau(T)]\right]\end{aligned}$$

# Non-Reneging Constraints

**Non-Reneging constraints**

$$\Pi_t \geq R_t \pi_L(q_L^*) + (1 - R_t)\pi_H(q_H^*), \quad \forall t \tag{NR}$$

If the principal were to renege, the worst punishment would be to cut off all future information.

- The best the principal can do after reneging is to
  - Extract max surplus from revealed low types;
  - Treat the unrevealed as high types (Assumption 1).
- The principal can guarantee a payoff arbitrarily close to the RHS.

# Incentive Compatibility

**IC-L**

$$U^S_{L,t}(t) = \sup_{T \geq t}(1-\delta)\sum_{\tau=t}^{T}\delta^{\tau-t}[x^P_\tau - C_L(q^P_\tau)] + \delta^T U^S_{L,T}(T), \quad \forall t \in \{t : 0 < r_t < \alpha - R_t\}$$

$$U^S_{L,t}(t) \geq \sup_{T \geq t}(1-\delta)\sum_{\tau=t}^{T}\delta^{\tau-t}[x^P_\tau - C_L(q^P_\tau)] + \delta^T U^S_{L,T}(T), \quad \forall t \in \{t : r_t \geq \alpha - R_t\} \tag{IC-L}$$

If some but not all low types reveal at $t$, then any low-type agent is indifferent.

**IC-H**

$$U^P_{H,t} \geq U^S_{H,t} := \sup_{T \geq t}(1-\delta)\sum_{\tau=t}^{T}\delta^{\tau-t}[x^S_\tau(t) - C_H(q^S_\tau(t))], \quad \forall t \tag{IC-H}$$

# IR Constraints and Implementable Allocations

**IR**

$$U^P_{H,t} \geq 0, \quad \forall t \tag{IR-H}$$

$$U^S_{L,t+s} \geq 0, \qquad \forall t \in \{t : r_t > 0\},\ \forall s \tag{IR-L}$$

**Definition 1**

*An allocation $\{(q^P_t, x^P_t), (q^S_\tau(T), x^S_\tau(T)), r_t\}_{t\geq 0, \tau \geq t}$ is **implementable** by a sequential revelation mechanism if it satisfies (NR), (IC-H), (IC-L), (IR-H), and (IR-L).*

# The Principal's Problem

The principal maximizes profit subject to the implementability constraints:

$$\sup_{\{(q_t^P, x_t^P), (q_\tau^S(T), x_\tau^S(T)), r_t\}_{t \geq 0, \tau \geq t}} \Pi_0$$

s.t. (NR), (IC-L), (IC-H), (IR-L), and (IR-H).

**Theorem 1 (Theorem 1)**

*There is a solution to the problem above. That is, an optimal mechanism exists.*

# Proof of Theorem 1

**Proof**

# Definitions

**Definition 2**

*A sequential revelation mechanism is **revealing** if $r_t > 0$ for some $t$. It is **non-revealing** if $r_t = 0$ for all $t$.*

**Definition 3**

*For a revealing mechanism, we say that information revelation is **gradual** if $r_t > 0$ for all $t$, and that information revelation is **eventual** if $\sum_{t=0}^{\infty} r_t = \alpha_0$.*

Assumptions:

- The max profit exceeds $\pi_H(q_H^*)$ (non-revealing mechanisms suboptimal).
- $\delta$ is sufficiently large.

# Frontloading of Payment

**Lemma 1 (Frontloading of Payment/Rent to L)**

*In any optimal mechanism, the separating contracts* $\{(q_\tau^S(t), x_\tau^S(t))\}_{\tau \geq t}$ *for agents revealing at t satisfy:*

$$x_\tau^S(t) = \begin{cases} C_H(q_\tau^S(t)), & \text{if } t \leq \tau < t+s \\ C_L(q_\tau^S(t)) + \beta \Delta C(q_\tau^S(t)), & \text{if } \tau = t+s \\ C_L(q_\tau^S(t)), & \text{if } \tau > t+s \end{cases}$$

*for some* $s \geq 1$ *and* $\beta \in (0, 1]$.

The separating contracts entail two phases:

1. Reward phase (debt periods): The principal pays a positive rent to a low-type agent for at least two periods (continuation payoff positive).
2. Exploitation phase: After $s+1$ periods, rent disappears forever.

# Logic Behind Frontloading

Trade-off

- Benefit: The principal can keep an agent's present value while frontloading rent.
  - This relaxes (NR) in future periods and allows strict profit improvement.
- Constraint: If she pays too much initially, the high type would pretend to be low, reap one-period high payment, and quit.
  - In any optimal mechanism, the high-type payoff is zero.
  - Maximum payment to low type is $C_H(q_\tau^S(t))$ (rent $\Delta C(q_\tau^S(t))$).

When $\delta$ is large, at least two debt periods are required to deliver the total rent.

# Pattern of Revelation

**Lemma 2 (Gradual and Eventual Information Revelation)**

*In any optimal mechanism, information revelation is gradual and eventual.*

## Gradual

- No final period: Otherwise, principal would stop paying rent after that period, contradicting Lemma 1 (at least two debt periods).
- No gaps: If $r_t, r_{t+2} > 0$ but $r_{t+1} = 0$, (NR) is slack at $t + 2$.
  - If she were to renege, she would rather renege at $t + 1$.
  - She will benefit from shifting agents who would reveal at $t + 2$ to $t + 1$.

## Eventual

- If $\sum_{t=0}^{\infty} r_t < \alpha_0$, scaling up $r_t$ in every period is feasible and reveals more information without violating constraints.

# Pooling Contracts

**Lemma 3 (Downward Distortion of H's Quantity)**

*In any optimal mechanism, the pooling contracts* $\{(q_t^P, x_t^P)\}_{t\geq 0}$ *satisfy* $q_t^P < q_H^*$ *for all* $t$ *and* $\lim_{t\to\infty} q_t^P = q_H^*$. *Moreover,* $x_t^P = C_H(q_t^P)$.

Insights

- Downward distortion: Analogue of standard screening distortion
- Zero payoff always for H
- Incentives: Lowering $q_t^P$ from $q_H^*$ strictly reduces rent for low types (positive payoff $\Delta C(q_t^P)$), with only 2nd-order effect on high-type profit.
- Convergence: Follows from Lemma 2 (all low types eventually opt out)

# Separating Contracts Quantities

**Lemma 4 (Upward Distortion and Frontloading of L's Quantity)**

*In any optimal mechanism, for agents revealing at t: If $\{t, \ldots, t+s\}$ are debt periods, then*

$$q_t^S(t) > q_{t+1}^S(t) > \cdots > q_{t+s}^S(t) \geq q_L^*$$

*and $q_\tau^S(t) = q_L^*$ for all $\tau > t+s$.*

- Upward distortion: Increasing $q$ above $q_L^*$ in debt periods helps frontload rent and relax future (NR).
- Decreasing Quantities
  - If $q_\tau^S(t) < q_{\tau+1}^S(t)$ for a debt period $\tau$, she can increase $(q_\tau^S(t), x_\tau^S(t))$ by $(\varepsilon, \varepsilon C_H')$ and decrease $(q_{\tau+1}^S(t), x_{\tau+1}^S(t))$ by $(\delta^{-1}\varepsilon, \delta^{-1}\varepsilon C_H')$.
  - This increases average surplus (concavity) and frontloads rent.

# Characterization of Optimal Mechanisms

**Theorem 2**

*Suppose non-revealing mechanisms are suboptimal. There exists a $\overline{\delta}$ such that if $\delta > \overline{\delta}$, then in every optimal mechanism:*

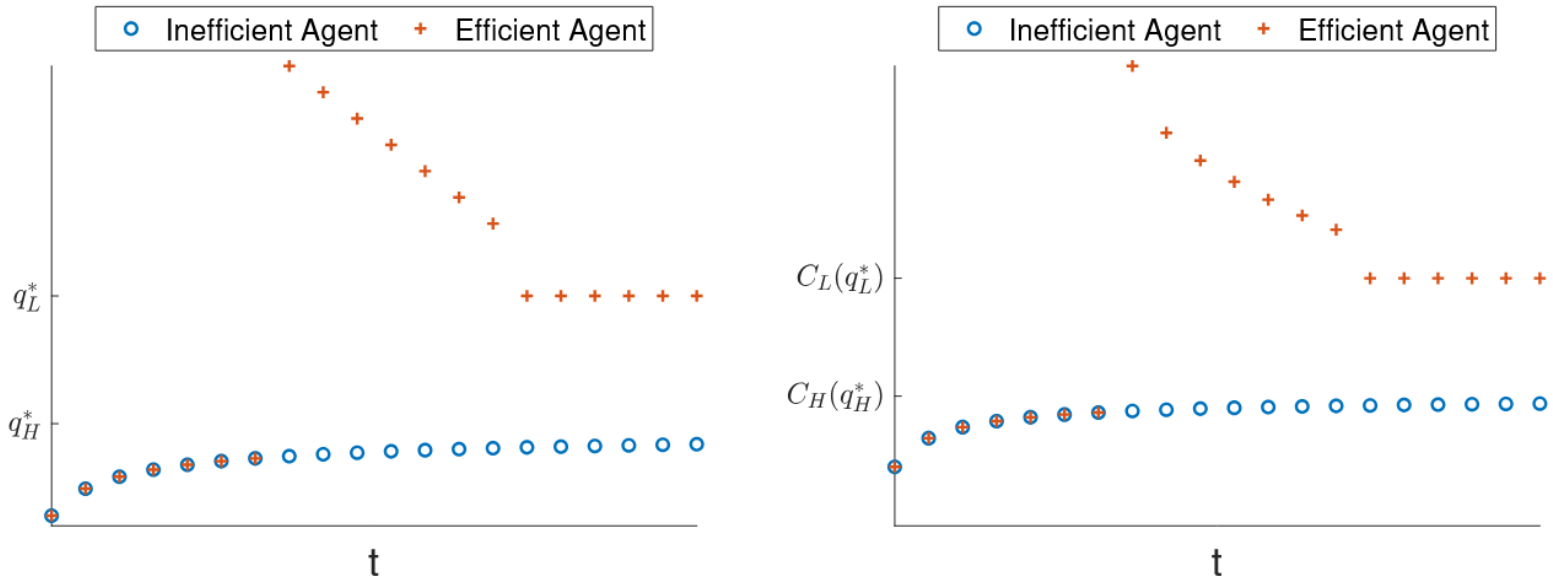
(i) *Information revelation is gradual and eventual.*

(ii) *After revelation, both amount of work and payment start high and decrease over time.*

(iii) *A low-type agent receives positive rents for a finite number of periods, then the principal extracts full surplus.*

(iv) *The amount of work in pooling contracts starts below the surplus-maximizing level and converges to it.*

# Life Cycle of an Agent's Contract




(a) Type-Dependent Production Dynamics (b) Type-Dependent Payment Dynamics

Figure 1: Life Cycle of an Agent's Contract. This figure is plotted under the following functional forms and parameter values: $v(q) = 2q^{1/2}$, $C_\theta(q) = \theta q$, $\theta_H = 3$, $\theta_L = 2$; the separating contracts are for an efficient agent who reveals his type at $t = 8$.

# Life Cycle of an Agent's Contract

Inefficient (high-cost) agent

- Wage and work amount gradually increase over time.
- Both remain at a relatively low level.

Efficient (low-cost) agent

- Contract exhibits an inverse U-shaped pattern.
- Promotion: Wage/work start low, but major promotion occurs once competence is learned.
- Exploitation: Due to limited commitment, rent is temporary. Payments and quantities gradually fall to surplus-maximizing levels.

# Information Revelation

**Theorem 3**

(i) *If for all* $\tilde{q}_L \in [q_L^*, \bar{q}]$,

$$\pi_L(q_L^*) - \pi_H(q_H^*) < \ln\left(\frac{\Delta C(\tilde{q}_L)}{\Delta C(\tilde{q}_L) - \Delta C(q_H^*)}\right)[\pi_L(q_L^*) - \pi_H(\tilde{q}_L)],$$

*then the (unique) optimal mechanism is non-revealing whenever* $\delta$ *is sufficiently close to 1.*

(ii) *If for some* $\tilde{q}_L \in [q_L^*, \bar{q}]$,

$$\pi_L(q_L^*) - \pi_H(q_H^*) > \ln\left(\frac{\Delta C(\tilde{q}_L)}{\Delta C(\tilde{q}_L) - \Delta C(q_H^*)}\right)[\pi_L(q_L^*) - \pi_H(\tilde{q}_L)],$$

*then every optimal mechanism is revealing whenever* $\delta$ *is sufficiently close to 1.*

# Interpretation of Theorem 3

Trade-off

- LHS: Maximum gains in surplus after learning agent type (Social value of information)
- RHS: Costs to induce separation
  - $\pi_L(q_L^*) - \pi_H(\tilde{q}_L)$: Profit loss per agent per debt period
  - $\ln(\dots)$: Accounts for the discounted number of debt periods needed to deliver rent

Every optimal mechanism is revealing $\Leftrightarrow$ The social value of information exceeds the incentivization costs.

# Intuitive Derivation

Consider a stationary auxiliary environment where

1. Constant measure $r$ reveal each period.
2. No distortion for high-cost agents.
3. For $T$ debt periods after revelation, principal demands $\tilde{q}_L$ and pays $C_H(\tilde{q}_L)$.

Reneging is never optimal $\Leftrightarrow$ Reneging one period *later* is better than *now*.

- Benefit: Additional measure $r$ reveal. Extra profit:

$$r\delta[\pi_L(q_L^*) - \pi_H(q_H^*)]$$

- Cost: Paying rent to agents revealed at $t-1, \ldots, t-T$.

$$(1-\delta)Tr[\pi_L(q_L^*) - \pi_H(\tilde{q}_L)]$$

# Intuitive Derivation (Endogenous Debt Periods)

The number of debt periods $T$ is determined by total information rent.

$$(1-\delta)\sum_{\tau=0}^{T-1}\delta^{\tau}\Delta C(\tilde{q}_L) = \Delta C(q_H^*)$$

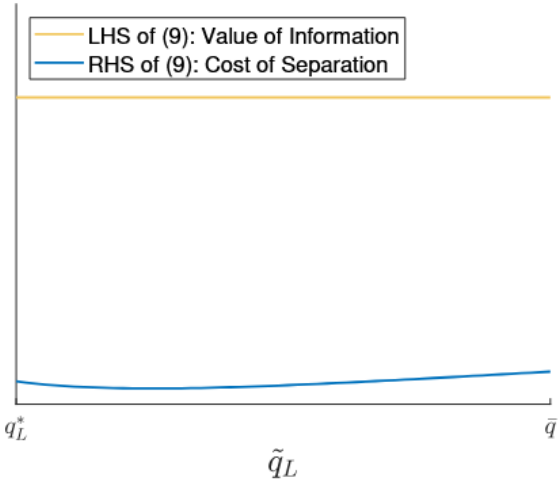
where

- RHS: Total rent needed (payoff if taking the pooling contract forever)
- LHS: Discounted sum of rents from $T$ debt periods

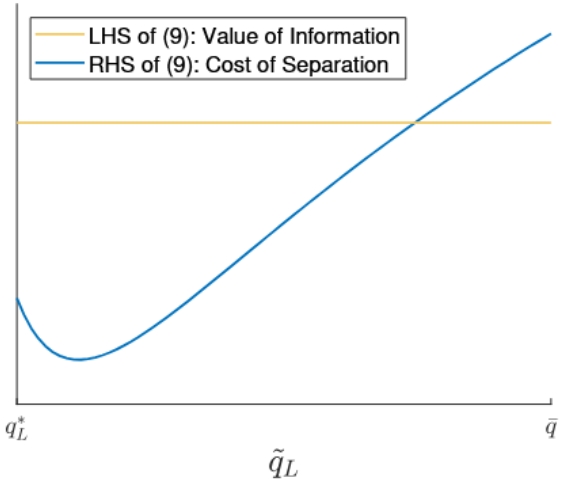
This implies

$$(1-\delta)T \to \ln\left(\frac{\Delta C(\tilde{q}_L)}{\Delta C(\tilde{q}_L) - \Delta C(q_H^*)}\right) \quad (\delta \to 1^-)$$

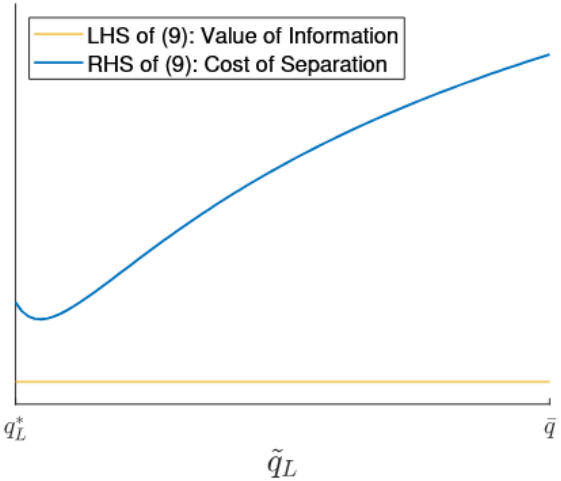
Combining benefit vs. cost yields the condition in Theorem 3.




(a) $\theta_L = 1.5$, $\theta_H = 3$ (Revealing)




(b) $\theta_L = 2$, $\theta_H = 3$ (Revealing)




(c) $\theta_L = 2.5$, $\theta_H = 3$ (Non-Revealing)

Figure 2: Necessary and Sufficient Condition for Information Revelation. This figure is plotted under the following functional forms and parameter values: $v(q) = 2q^{1/2}$, $C_\theta(q) = \theta q$, $\bar{q} = 1$, $\theta_H = 3$, $\theta_L = 1.5$ for (a), $\theta_L = 2$ for (b), and $\theta_L = 2.5$ for (c).

# Discussion: Cardinality of Agents

There exists an environment with countably many agents in which information revelation is feasible.

- Finitely many agents per period, with replacement (new arrivals).

For a fixed pool of finitely many agents,

- Unique equilibrium outcome (for patient players) is pooling.
  1. If revealing, there must be a final period of revelation.
  2. After that, principal has no future info to learn/lose $\Rightarrow$ full exploitation.
  3. Anticipating this, no agent reveals in the final period.

# Literature Review

The ratchet effect

- Freixas et al. (1985), Laffont and Tirole (1988, 1993)
- Infinite time horizon: Gerardi and Maestri (2020)
- Institutions that help mitigate the ratchet effect: Kanemoto and MacLeod (1992), Carmichael and MacLeod (2000), Acharya and Ortner (2017).

Limited commitment...

Interacting with multiple agents help the principal keep her promises.

- Moral-hazard with complete information: Levin (2002)
- Selling a durable good to a continuum of consumers: Ausubel and Deneckere (1989)

# Takeaway

- This paper shows that a dynamic contract with multiple agents and one-period commitment can mitigate the ratchet effect.
  - Information revelation is gradual and eventual.
  - An efficient agent's contract exhibits an inverse U-shaped pattern.
- Commitment power is partially restored with multiple agents and observable reneging.
- Revelation is feasible if and only if the social value of information exceeds the cost of incentivizing separation.